

WEB TECHNOLOGIES

CSE-4004



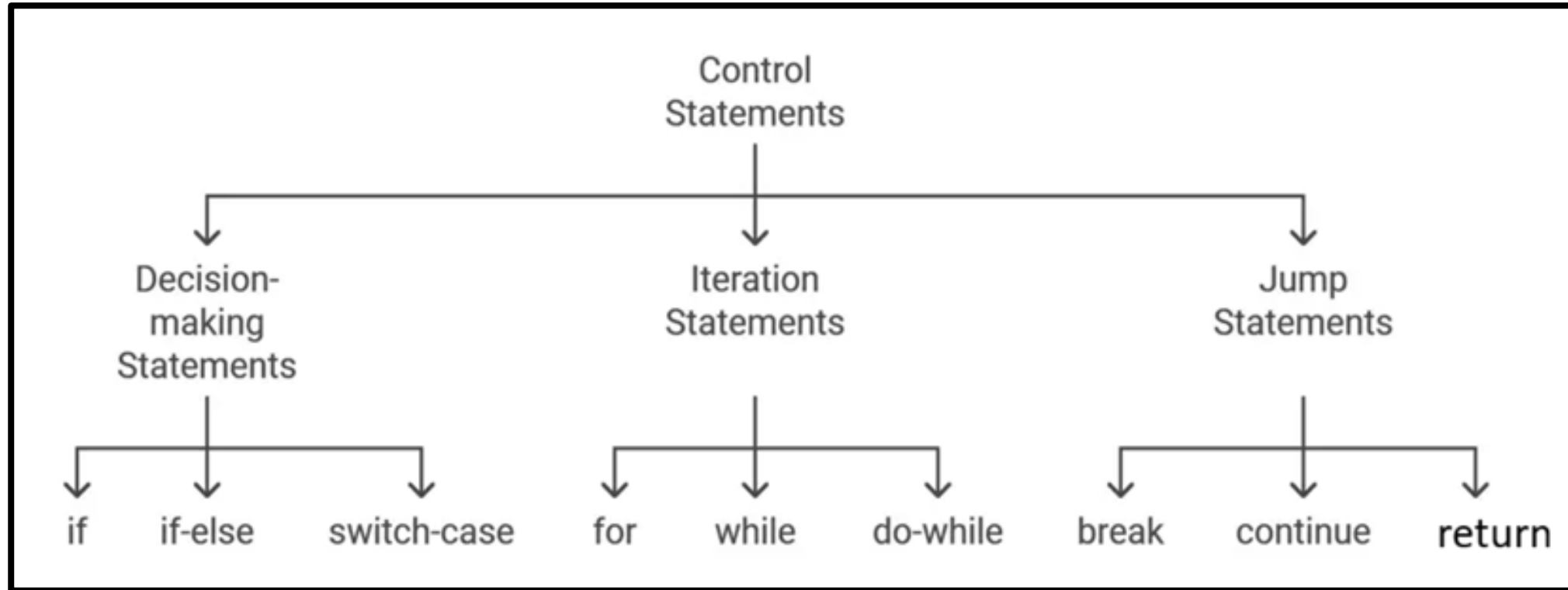
LECTURE 2: JS Control Structures and Programming



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Control Structure

- Controls the flow of execution of a program.



Control Structure – Decision Making Statement

- **Execute instructions based on the given condition**

```
if (condition)  
{  
    // code  
}
```

```
if (condition)  
{  
    // true block  
}  
else  
{  
    // false block  
}
```

Test Multiple conditions

```
if (cond1)  
{  
    // code  
}  
else if (cond2)  
{  
    // code  
}  
else  
{  
    // code  
}
```

```
switch (expression)  
{  
    case value1: stmt 1;  
        break;  
    case value2: stmt 2;  
        break;  
  
    default: default stmt;  
}
```

Control Structure – Looping Statement

- **Execute instructions repeatedly based on a condition**

```
for (initialization; condition; increment)
{
    // body of the loop
}
```

```
while (condition)
{
    // body of the loop
}
```

```
do
{
    // body of the loop
} while (condition);
```

```
for (let value of iterable)
{
    // code using value
}
```

Example:

```
let a=[10,20];
for(let x of a)
{
    document.write(x + "<br>");
}
```

Output:

```
10
20
```

Control Structure – Looping Statement

- **Execute instructions repeatedly based on a condition**

Iterates over the enumerable properties of an object

for (let variable in iterable)

```
{  
  // code  
}
```

Example:

```
let a=[10,20];  
for(let x in a)  
{  
  document.write(a[x] + "<br>");  
}
```

Output:

10

20

Control Structure – Jumping Statement

- Transfers the control

break: terminates loop or switch statement

continue: statement skips the current iteration and moves to the next iteration of the loop

return: statement exits from a function and optionally returns a value.

Simple Calculator

// Complete Calculator using prompt, alert, and switch

```
let iterate = true;
while(iterate)
{
    let num1 = parseInt(prompt("Enter the first number:"));
    let num2 = parseInt(prompt("Enter the second number:"));
    let operator = prompt("Enter the operation (+, -, *, /, %):");
    let result;
    switch (operator) {
        case "+": result = num1 + num2;
                  break;
        case "-": result = num1 - num2;
                  break;
        case "*": result = num1 * num2;
                  break;
```

Simple Calculator

```
case "/": if (num2 === 0)
    {
        alert("Error: Division by zero!");
        continue;
    }
    result = num1 / num2;
    break;
case "%": if (num2 === 0)
    {
        alert("Error: Division by zero!");
        continue;
    }
    result = num1 % num2;
    break;
```

Simple Calculator

```
default: alert("Invalid operator!");  
        continue;  
        }  
  
alert("Result: " + result);  
let again = prompt("Do you want to continue? (y/n)");  
if (again !== "y")  
{  
    iterate = false;  
    alert("Thank you for using the calculator!");  
}  
}
```

OUTPUT

ntp.msn.com says

Enter the first number:

ntp.msn.com says

Enter the operation (+, -, *, /, %):

ntp.msn.com says

Enter the second number:

ntp.msn.com says

Result: 0.5

OUTPUT

ntp.msn.com says

Thank you for using the calculator!

OK

ntp.msn.com says

Do you want to continue? (y/n)

n

OK

Cancel